

ECOMMERCE SALES ANALYTICS BY SQL QUERIES IN MYSQL RDBMS

-- Total Revenue

```
select sum(revenue) from ec;
```

sum(revenue)
5109775.740000007

-- Total Number of Orders

```
select count(*) from ec;
```

count(*)
5000

-- Total Number of Unique Customers

```
select count( distinct customer_id) as 'Total Number of Unique Customers' from ec;
```

Total Number of Unique Customers
989

-- Average Customer Rating

```
select round(avg(customer_rating),2) 'Average Customer Rating' from ec;
```

Average Customer Rating
2.97

-- Average Delivery Days

```
select avg(delivery_days) 'Average Delivery Days' from ec;
```

Average Delivery Days
6.1188

-- Monthly Revenue Trend

```
select month(order_date) as month, round(sum(revenue),2) 'Total Revenue'  
from ec  
group by month(order_date);
```

month	Total Revenue
1	427937.77
2	378784.31
3	428440.9
4	415090.15
5	460351.11
6	426291.61
7	434202.45
8	476059.4
9	407503.08
10	415573.62
11	410572.67
12	428968.67

-- Revenue by Product Category

```
select product_category, round(sum(revenue),2) 'Total Revenue'  
from ec  
group by product_category;
```

product_category	Total Revenue
Beauty	765860.88
Clothing	1531931.72
Electronics	1829899.22
Home	982083.92

-- Revenue by Region

```
select region, round(sum(revenue),2) 'Total Revenue'
```

from ec

group by region;

region	Total Revenue
South	1246640.9
East	1236044.23
West	1345582.16
North	1281508.45

-- Revenue by Payment Method

select payment_method, round(sum(revenue),2) 'Total Revenue'

from ec

group by payment_method;

payment_method	Total Revenue
Wallet	955119.1
Card	2366248.2
COD	1788408.44

-- Top 5 Highest Paying Customers

select customer_id, round(sum(revenue),2) 'Total Revenue'

from ec group by customer_id

order by sum(revenue) desc limit 5;

customer_id	Total Revenue
1663	17680.69
1955	16265.15
1675	15036.46
1276	15023.71
1647	14894.97

-- Top 5 Customers by Quantity Purchased

select customer_id, sum(quantity) Total_orders

from ec group by customer_id

order by Total_orders desc limit 5;

customer_id	Total_orders
1663	63
1957	57
1675	55
1935	51
1276	50

-- Top 5 Highest Rated Customers

```
select *, dense_rank() over (order by avg_rating desc)
from (select customer_id, round(avg(customer_rating),2) avg_rating
from ec
group by customer_id) as ar
limit 5;
```

customer_id	avg_rating	dense_rank() over (order by avg_rating desc)
1594	4.9	1
1990	4.83	2
1443	4.8	3
1049	4.75	4
1650	4.75	4

-- Repeat Customers

```
select customer_id ,count(customer_id) as count_of_customer
from ec group by customer_id
having count_of_customer >1;
```

customer_id	count_of_customer
1102	5
1435	7
1860	5
1270	4
1106	5
1071	4
1700	6
1020	7
1614	3
1121	6
1466	7
1214	4
1330	3
1458	7
1087	4
1372	5
1099	3
1871	8
1663	14
1130	8
1661	9
1308	4

-- Average Rating by Product Category

```
select product_category, round(avg(customer_rating),3) 'avg rating'
from ec group by product_category;
```

product_category	avg rating
Beauty	3.009
Clothing	3.005
Electronics	2.951
Home	2.941

-- Orders by Product Category (%)

```
select product_category, orders_per_category *100/ (select count(*) from ec) '%
contribution' from
```

```
(select product_category, count(*) as orders_per_category
```

```
from ec group by product_category) as t;
```

product_category	% contribution
Beauty	14.4600
Clothing	30.6200
Electronics	35.5400
Home	19.3800

-- Percentage of Orders by Payment Method

```
select payment_method, orders_per_category *100/ (select count(*) from ec) '%
contribution'
```

```
from (select payment_method, count(*) as orders_per_category
```

```
from ec group by payment_method) t;
```

payment_method	% contribution
Wallet	19.1200
Card	45.4000
COD	35.4800

-- Rank Regions by Revenue

```
select *, dense_rank() over(order by total_revenue desc) as region_rank from  
(select region, round(sum(revenue),2) as total_revenue from ec group by region) t;
```

region	total_revenue	region_rank
West	1345582.16	1
North	1281508.45	2
South	1246640.9	3
East	1236044.23	4

-- Monthly Revenue Growth

```
select month_name,  
current_month_revenue ,  
round((current_month_revenue-previous_month_revenue)*100/ previous_month_revenue,  
2)as 'monthly revenue growth' from (select *, lag(current_month_revenue) over(order by  
month_no) previous_month_revenue  
from (select month(order_date) as month_no, monthname(order_date) as month_name,  
round(sum(revenue)) as current_month_revenue  
from ec group by month_no, month_name) t) tt
```

month_name	current_month_revenue	monthly revenue growth
January	427938	NULL
February	378784	-11.49
March	428441	13.11
April	415090	-3.12
May	460351	10.9
June	426292	-7.4
July	434202	1.86
August	476059	9.64
September	407503	-14.4
October	415574	1.98
November	410573	-1.2
December	428969	4.48